

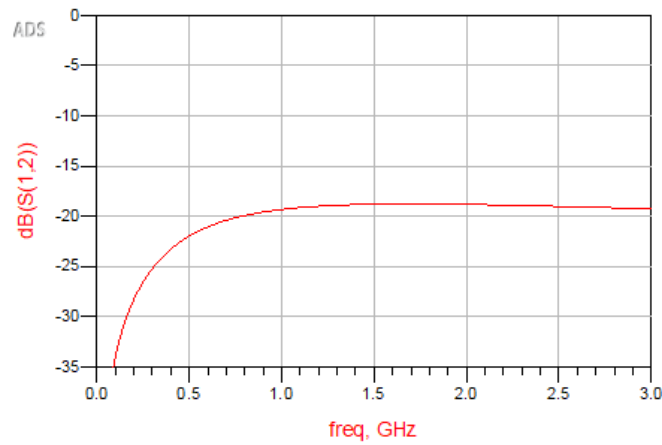
RF Design Course 2026

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1 Task10

1.1 Number 4



The reverse isolation S_{12} is very small at low frequencies, indicating weak feedback from output to input. As frequency increases, the parasitic transistor capacitances provide a coupling path between output and input.

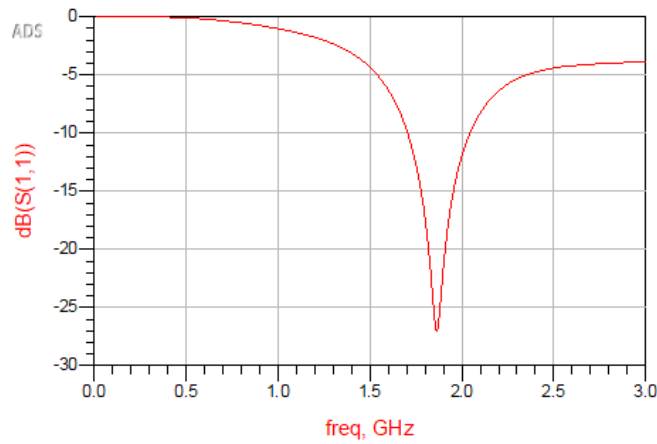
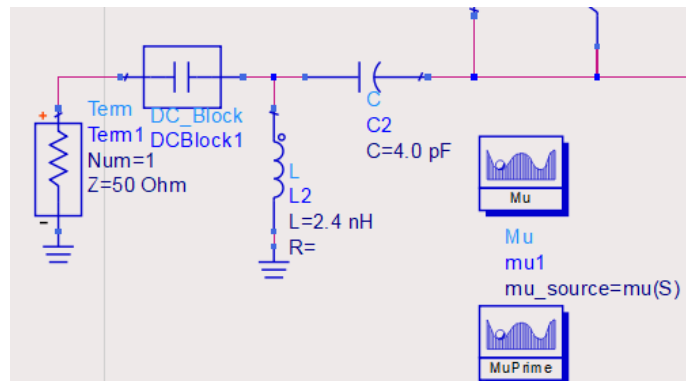
1.2 Number 5

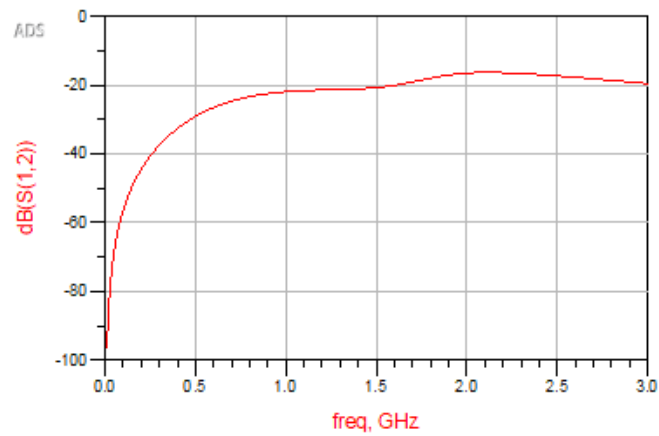
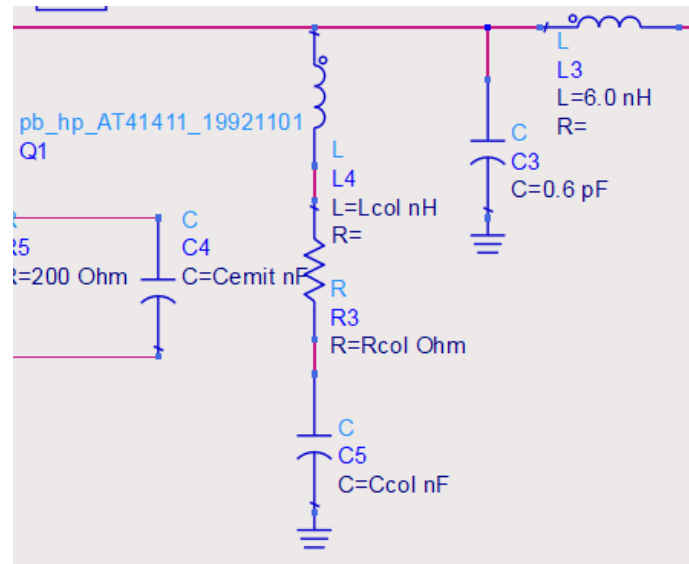
The feedback is influenced not only by the transistor collector-base capacitance C_{bc} , but also by the external collector-base resistor R_4 . Furthermore, the emitter resistor R_5 introduces negative feedback, while the bypass capacitor $C1$ reduces this feedback at RF frequencies.

1.3 Number 6

Table 1: Functions of selected networks in the amplifier

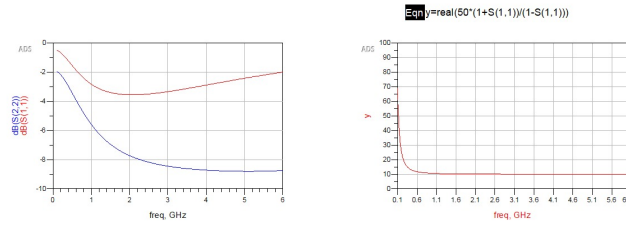
Network	Components	Main Function
Input Matching Network	L_2, C_2	Transforms the $50\ \Omega$ source impedance at the input. Its main purpose is to reduce the input reflection coefficient S_{11} , while it also influences gain and stability through the source impedance seen by the transistor.
Collector-to-base Feedback	R_2, L_1, C_1	Provides a controlled feedback path from the collector side to the input. The feedback is frequency-dependent and helps to shape the frequency response and improve stability.





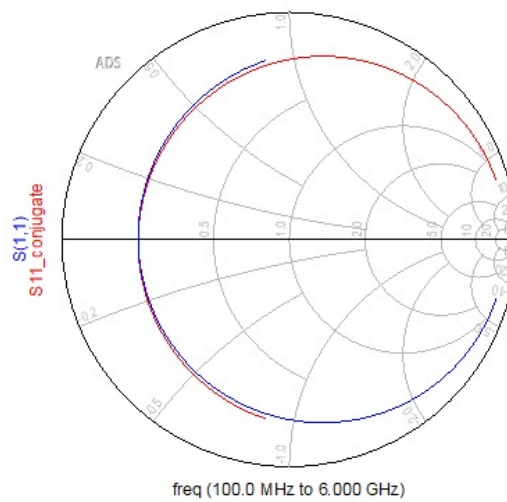
1.4 Number 7

The plot of the real part of the impedance for frequency varied from 100 MHz... 6 GHz is as shown:



1.5 Number8

$$\text{Eqn } S_{11_conjugate} = \text{conj}(S(1,1))$$

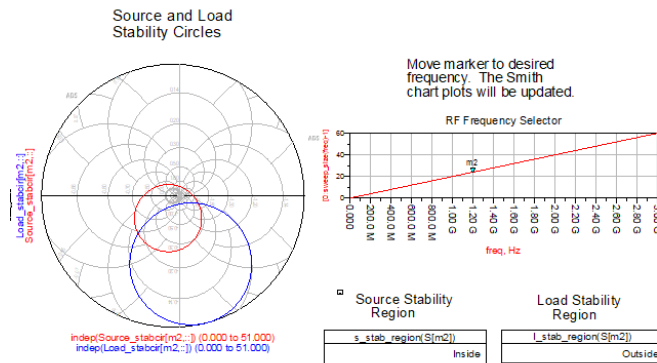


1.6 Number9

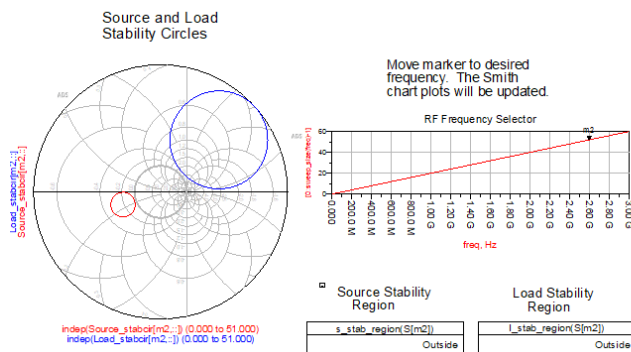
The required conjugate matching impedance changes continuously with frequency. A finite lumped LC network cannot exactly realize this frequency-dependent impedance over the whole range from 100 MHz to 6 GHz.

1.7 Number10

Set VDC=2V so VCE=1.2V
1.2Ghz

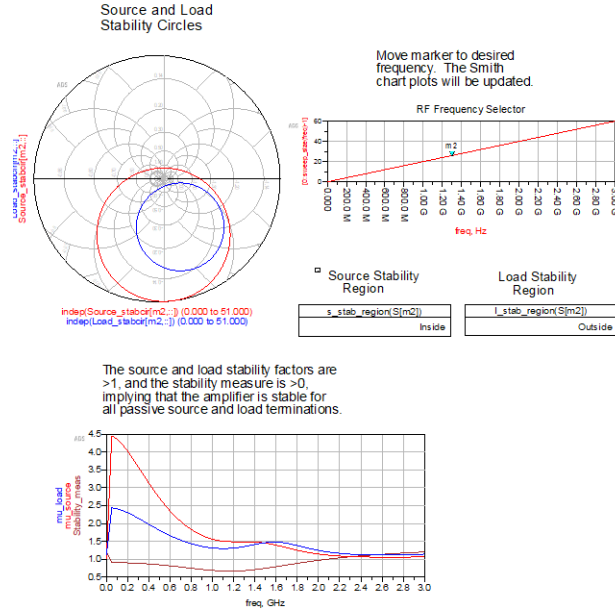


2.6Ghz



1.8 Number 11

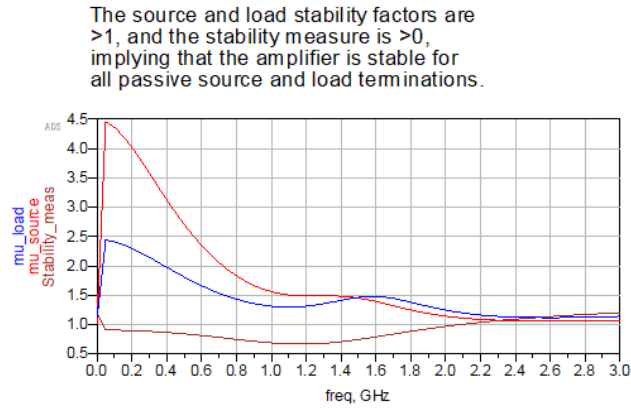
1.3Ghz is stable



When the collector damping resistor is changed to $R_{col} = 0 \Omega$, the stability factor drops below one, so the amplifier becomes unstable.

1.9 Number 12

The factors μ_{load} and μ_{source} describe how far the amplifier is from instability for passive load and source terminations. Values larger than one indicate unconditional stability. They are useful because they give a direct frequency-dependent stability margin instead of requiring manual interpretation of Smith-chart stability circles.



1.10 Number 13

The stability measure is a numerical indicator of the distance to instability. The factors μ_{source} and μ_{load} are easier to use because a value greater than one immediately indicates unconditional stability. Their advantages are simple interpretation, easy frequency-domain plotting, and no need for graphical analysis of stability circles and stable regions.

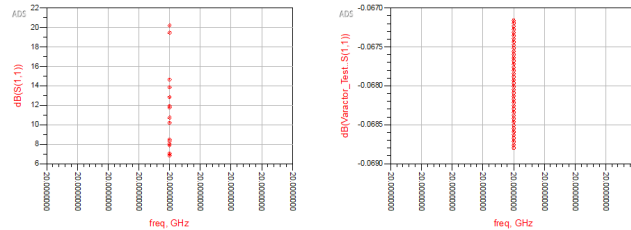
2 Task 11

2.1 Number 2

- The main electrical representation of a Varactor is a voltage controlled variable capacitance.
- The varactors are presetly implemented using MOSFETs(MOSCAPS).
- The Drain, Source and Bulk terminals are shorted to form one terminal and the Gate becomes the other terminal.

2.2 Number 3

- $\Gamma_{in} > 1$ occurs when the real part of impedance is negative.
- This negative resistance is achieved by biasing the transistor in the unstable region.
- The magnitude exceeding 1 ensures the loop gain condition for oscillation startup is satisfied.



2.3 Number 4

- In a 2-port oscillator configuration (like a FET with feedback):
 Γ_{in} = Reflection coefficient at the input port (looking into the active device)
 Γ_{out} = Reflection coefficient at the output port (looking into the feedback network)

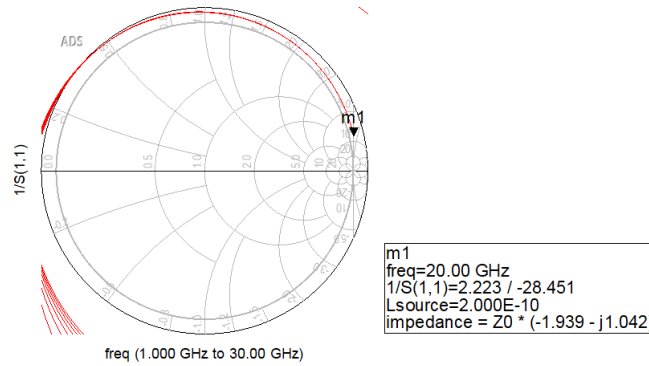
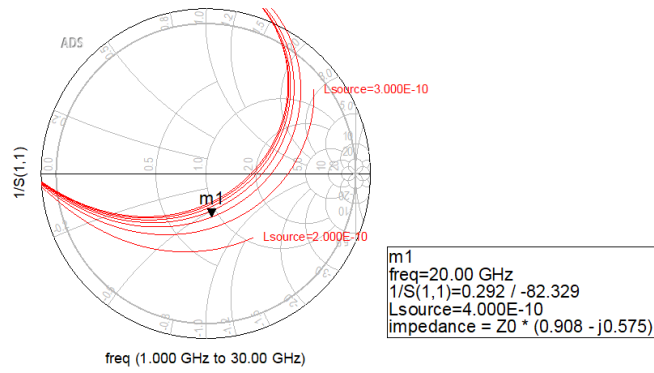
The loop gain is the product:

$$\text{Loop Gain} = \Gamma_1 \times \Gamma_2$$

When this product satisfies the two conditions above, the circuit will oscillate.

- For an oscillator to sustain stable oscillations, two conditions must be satisfied simultaneously:
Magnitude Condition - $|\Gamma_1 \times \Gamma_2| \geq 1$
Phase Condition - $\text{Angle}(\Gamma_1 \times \Gamma_2) = 0^\circ$
- Frequency and Phase relation: Oscillation occurs at the frequency where the total phase shift around the loop is exactly 0° (or multiples of 360°).

2.4 Number 5



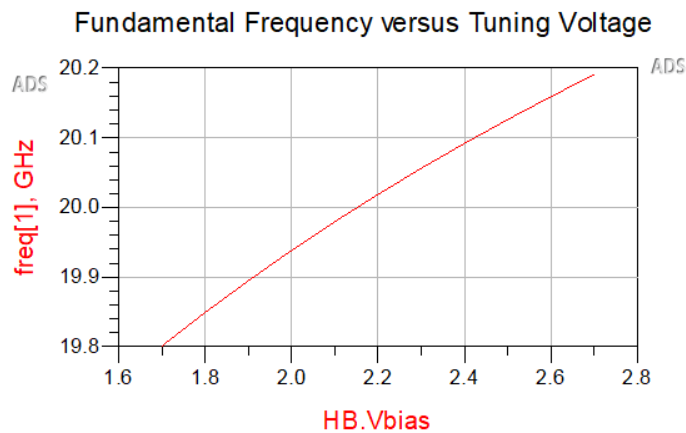
- At $V_{dc} = -0.31V$, the FET is in its active region with significant drain current, resulting in positive input resistance and $|\Gamma| < 1$ (stable).

- At $V_{dc} = -0.9V$ (near pinch-off), the drain current drops to nearly zero, the transconductance collapses, and the FET behaves like a capacitive network with feedback through C_{gd} .
- This feedback can generate negative resistance at the input, causing $|Gamma| > 1$ (instability) — a necessary condition for oscillation.
- Yes, a device in the off-state can be unstable because the feedback capacitances (C_{gd} , C_{gs}) can create a negative resistance even with zero DC current, which is exploited in many microwave oscillators.

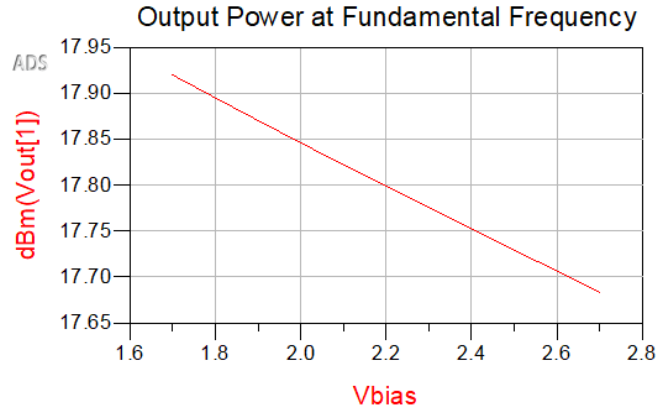
2.5 Number 7

- The varactor acts as a voltage-tunable capacitor in the resonator tank circuit.
- By changing the bias voltage V_{bias} , the varactor's junction capacitance changes, which shifts the resonant frequency of the LC tank.
- As V_{bias} increases (reverse bias increases), the depletion region widens, causing the varactor capacitance to decrease and the resonant frequency to increase.
- The defining quantity of a varactor is its capacitance-voltage (C-V) characteristic.
- The varactor enables voltage-controlled oscillation (VCO) by tuning the oscillator frequency with a DC bias voltage.
- $Tuningratio(orCapacitiveratio) = C_{max}/C_{min} = C(@V_{min})/C(@V_{max})$

2.6 Number 8

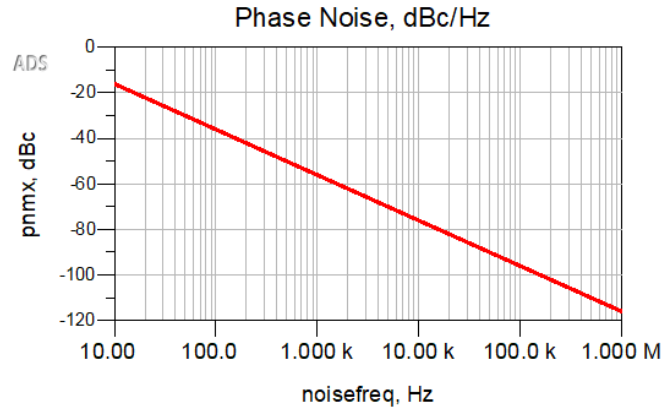


After reducing the oscillator core supply to 2.35 V, the tuning voltage V_{bias} must be increased to 2.2 V to restore the fundamental oscillation frequency to 20 GHz.



The output power at the fundamental frequency decreases monotonically as V_{bias} increases, from approximately 17.92 dBm at $V_{\text{bias}} = 1.7$ V to 17.68 dBm at $V_{\text{bias}} = 2.7$ V, due to increased varactor series resistance and reduced tank Q at higher reverse bias voltages.

2.7 Number 9



- To minimize phase noise at a 100 kHz offset, increase the quality factor (Q) of the resonator tank circuit, as phase noise is inversely proportional to Q^2 , which reduces the up-conversion of white noise.
- Select an active device (FET or BJT) with a lower flicker noise (1/f) corner

frequency, and ensure it is biased to operate in a region with minimal low-frequency noise up-conversion.

2.8 Number 10

- S12 is checked for a buffer amplifier because it represents the reverse isolation — how much of the output signal (or load reflections) leaks back to the input (the oscillator).
- The situation being considered is preventing frequency pulling of the oscillator caused by load impedance variations, such as when an antenna is detuned or a mixer presents different impedances at different LO drive levels.
- A low S12 ensures the oscillator sees a constant load impedance regardless of what is connected to the buffer output, maintaining stable oscillation frequency.

3 Task 12

3.1 Number 3

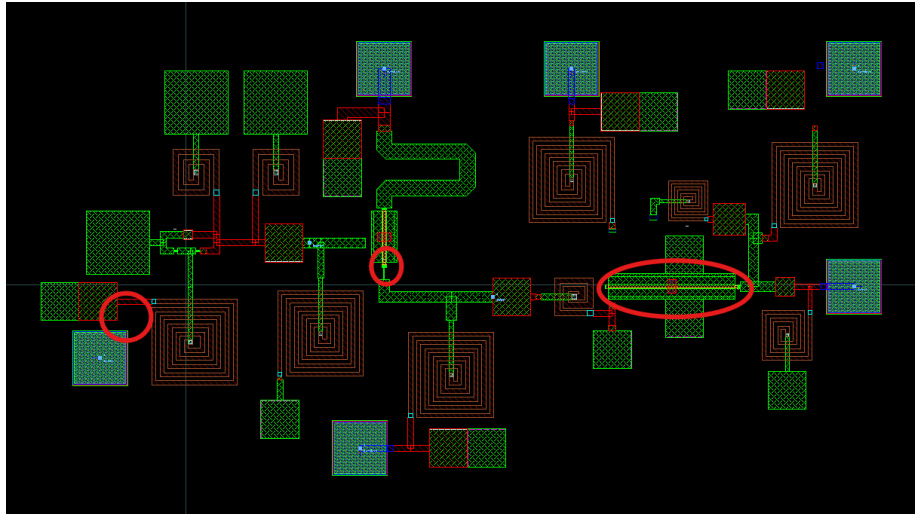
Table 2: Examples of rule violation in DRC and related errors

Rule Violation	Typical IC Error
Gate must be inside Metal ≥ 0	Gate metal misaligned or undersized, causing poor ohmic contact
Minimum metal spacing	Adjacent metal lines too close, risking short circuits
Minimum via size	Vias too small, leading to high resistance or open circuits
Active to well edge spacing	Active region too close to well boundary, causing latch-up risk
Metal density rule	Uneven metal coverage causing wafer warpage during manufacturing

3.2 Number 4

LVS (Layout Versus Schematic) compares the generated layout with the original schematic and verifies that all components and electrical connections are identical. It detects missing components, wrong device parameters, and connectivity errors between nets. Problems occur when a component is missing, incorrectly connected, or when the layout does not match the schematic.

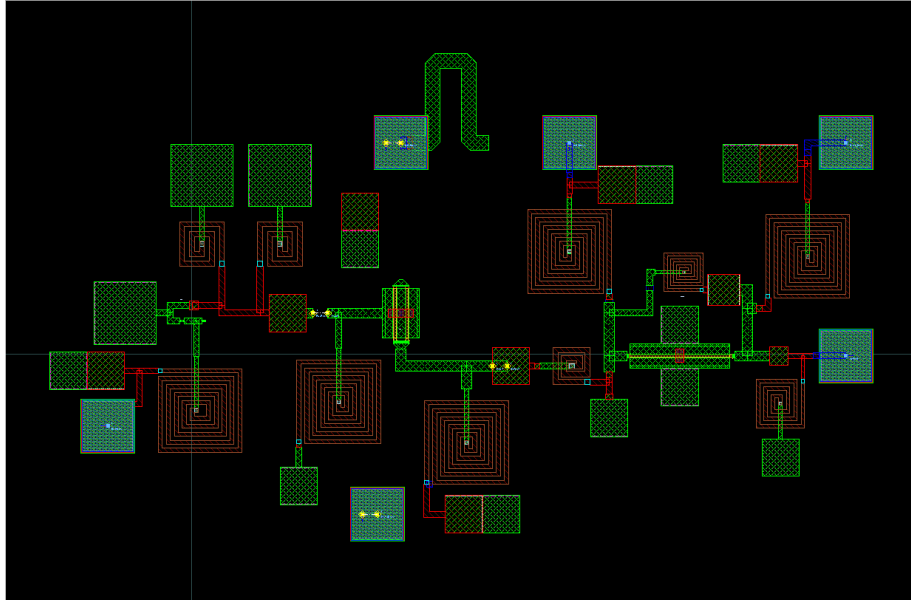
3.3 Number 5



Screenshot of Layout showing the three critical changes.

- We created three critical changes: (1) deleted a $Node_3$ trace creating an open circuit, (2) added a wire short between two nets, and (3) changed the FET gate width from 200 μm to 250 μm .
- The LVS detected nodal mismatches for the open circuits, a parameter mismatch for the FET width change, and a Warning for flat lines
- This demonstrates that LVS effectively catches both connectivity errors and device parameter discrepancies between schematic and layout.

3.4 Number 6



Screenshot of Layout showing changes after removal of buffer stage.

- We removed the BufferStage component from the CompleteOsc schematic and regenerated the layout.
- The LVS reported a "Components not in layout" error because the schematic had 1 fewer component than the layout.
- The design was not error-free — LVS correctly identified the mismatch, confirming that schematic and layout must have identical component instances.